CMPE 403: Computer Architecture & Organization

Chapter 3: Control Unit — Visual Guide & Complete Study Reviewer

VISUAL REVIEWER

1. CONTROL UNIT OVERVIEW & CORE CONCEPTS

Control Unit (CU) Nerve Center

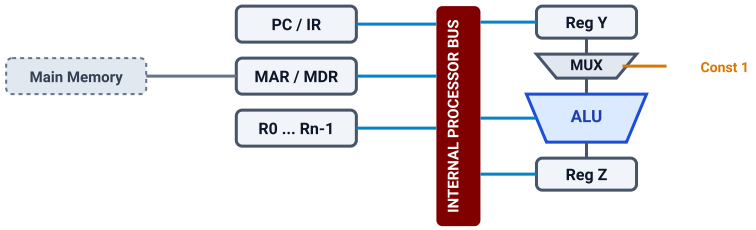
The CU coordinates all hardware operations. It fetches instructions from main memory, decodes them in the Instruction Register (IR), and generates precise timing/control signals to direct data flows between registers, ALU, and external memory without processing data itself.

Control Memory & Microprograms

Internal high-speed ROM storing microinstructions. Each machine instruction triggers a microprogram (fetch, evaluate address, execute). Writable Control Store (WCS) uses RAM to allow microcode modification.

2. SINGLE-BUS DATAPATH VISUAL ARCHITECTURE

FIGURE 3.1: INTERNAL PROCESSOR SINGLE-BUS DATAPATH & GATE CONTROLS



3. DATAPATH COMPONENTS & GATE SIGNALS

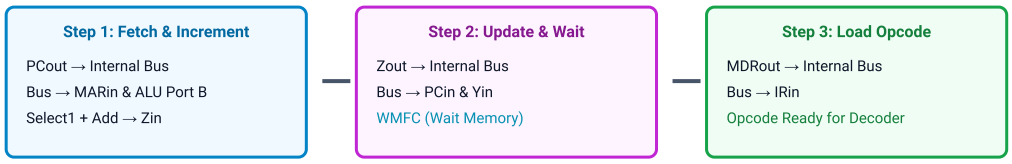
Component	Role & Function in Single-Bus Architecture	Control Gate Signals
PC	**Program Counter:** Holds the address of the next instruction to fetch.	`PCin`, `PCout`
MAR / MDR	**Memory Registers:** MAR sends address to memory; MDR holds incoming/outgoing data.	`MARin`, `MDRin`, `MDRout`, `MDRoutE`
IR	**Instruction Register:** Holds fetched opcode for Instruction Decoder processing.	`IRin`
Reg Y & Reg Z	**ALU Buffers:** Reg Y holds Input A; Reg Z holds output of ALU before bus transfer.	`Yin`, `Zin`, `Zout`

4. HOW CONTROL SEQUENCES WORK: RULES & VISUAL PATHS

THE 3 GOLDEN RULES OF CONTROL SEQUENCES:
1. **Single Output Rule:** Only ONE register can output to the bus at any cycle (e.g., `PCout` OR `R1out`). Multiple outputs cause bus collision!
2. **Multiple Input Rule:** Multiple registers can read from the bus at the same cycle (e.g., `MARin` and `Yin` together).
3. **ALU Routing Rule:** ALU output cannot go directly to general registers—it must be locked into `Reg Z` first (`Zin`) then outputted in the next cycle (`Zout`).

VISUALIZING DATA FLOW PATHS

FIGURE 3.2: PARALLEL SIGNAL COMBINATIONS VS. SEQUENTIAL CYCLES



5. UNIVERSAL INSTRUCTION FETCH SEQUENCE (STEPS 1–3)

STEP 1: Initiate Memory Read & Increment PC

`PCout, MARin, Read, Select1, Add, Zin`

Places PC onto bus into MAR; sends Read signal to memory bus. MUX selects Constant 1 (Port A) and PC from bus (Port B); ALU adds them and stores updated address (PC+1) into Reg Z.

STEP 2: Update PC, Copy to Y, and Wait for Memory

`Zout, PCin, Yin, WMFC`

Transfers updated address from Reg Z into PC and Reg Y (for potential branch offset calculation). Processor waits for Memory Function Complete (WMFC) signal.

STEP 3: Load Instruction into IR

`MDRout, IRin`

Transfers fetched instruction word from MDR into Instruction Register (IR) for decoding.

6. SOLVED EXERCISE PROBLEMS (STEP-BY-STEP)

EXERCISE 1, PROBLEM 1: R2 ← [R2] + [[R1]] — **Single Indirect Memory Read + Add**

```
Step 1: PCout, MARin, Read, Select1, Add, Zin
Step 2: Zout, PCin, Yin, WMFC
Step 3: MDRout, IRin
Step 4: R1out, MARin, Read
Step 5: R2out, Yin, WMFC
Step 6: MDRout, SelectY, Add, Zin
Step 7: Zout, R2in, End
```

Explanation: Steps 1–3 fetch instruction. Step 4 sends memory address in R1 to MAR and initiates Read. Step 5 moves R2 into Reg Y while waiting for memory (WMFC). Step 6 adds fetched data (MDR) to Reg Y. Step 7 saves sum into R2.

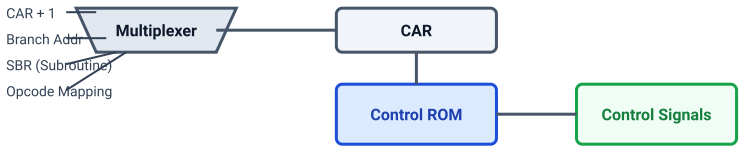
EXERCISE 1, PROBLEM 2: R1 ← [R1] + [[[R2]]] — **Double Indirect Pointer Chain + Add**

```
Step 1: PCout, MARin, Read, Select1, Add, Zin
Step 2: Zout, PCin, Yin, WMFC
Step 3: MDRout, IRin
Step 4: R2out, MARin, Read
Step 5: WMFC
Step 6: MDRout, MARin, Read
Step 7: R1out, Yin, WMFC
Step 8: MDRout, SelectY, Add, Zin
Step 9: Zout, R1in, End
```

Explanation: Step 4 reads 1st pointer from address in R2. Step 6 reads 2nd pointer using address received in MDR. Step 7 loads R1 into Reg Y while waiting for final data operand. Step 8 adds data operand to R1; Step 9 saves result into R1.

7. ADDRESS SEQUENCING ARCHITECTURE

FIGURE 3.3: MICROPROGRAM ADDRESS SEQUENCING LOGIC



8. ADDRESS SEQUENCING FEATURES & TERMS

Concept	Definition & Role in Control Unit
CAR	**Control Address Register:** Holds the address of the next microinstruction in Control ROM.
SBR	**Subroutine Register:** Saves return address when executing microprogram subroutines.
Opcode Mapping	Translates machine opcode from IR into starting address of corresponding microprogram.
Branch Logic	Evaluates status flags/bits to decide next microinstruction address (CAR+1 vs Branch Address).